

Vanishing Points and Cross Ratio

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Outline

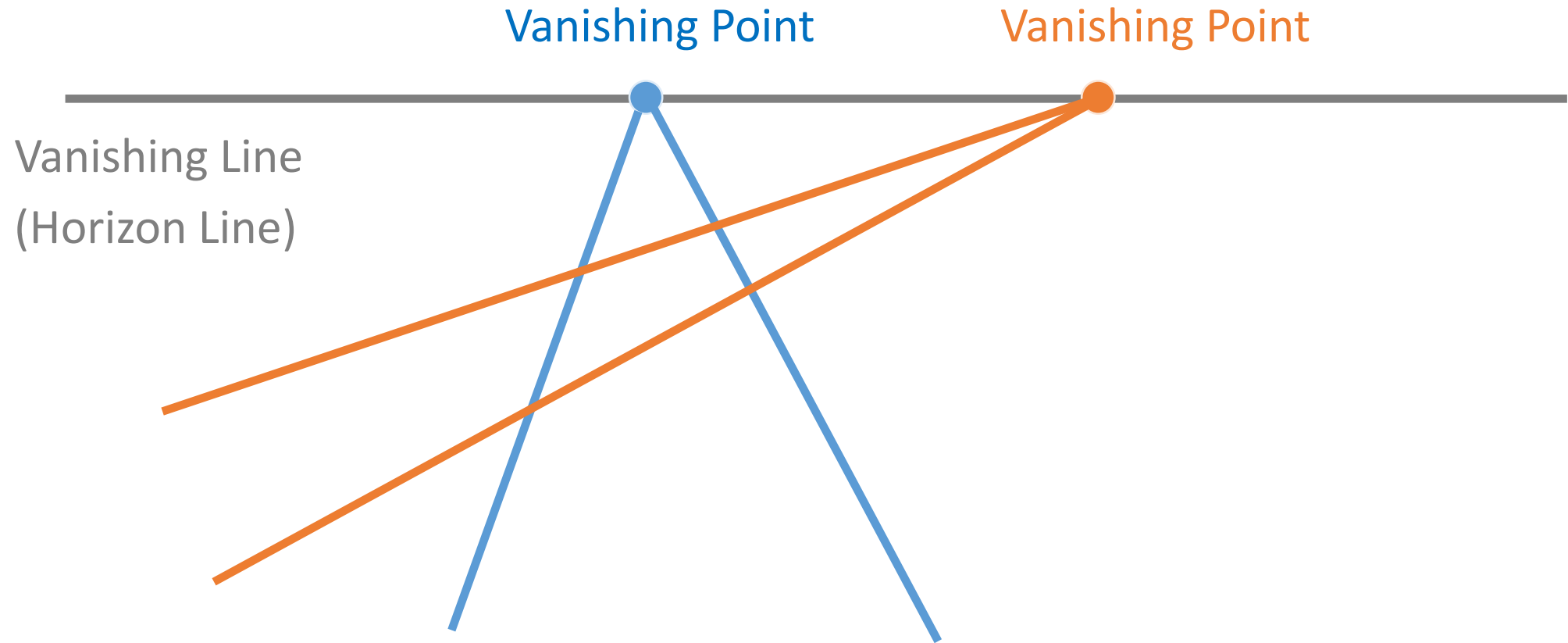
- Vanishing Points (VP) and Horizon Line (HL)
- Detecting VPs and HL
- Cross-Ratio

Vanishing Points

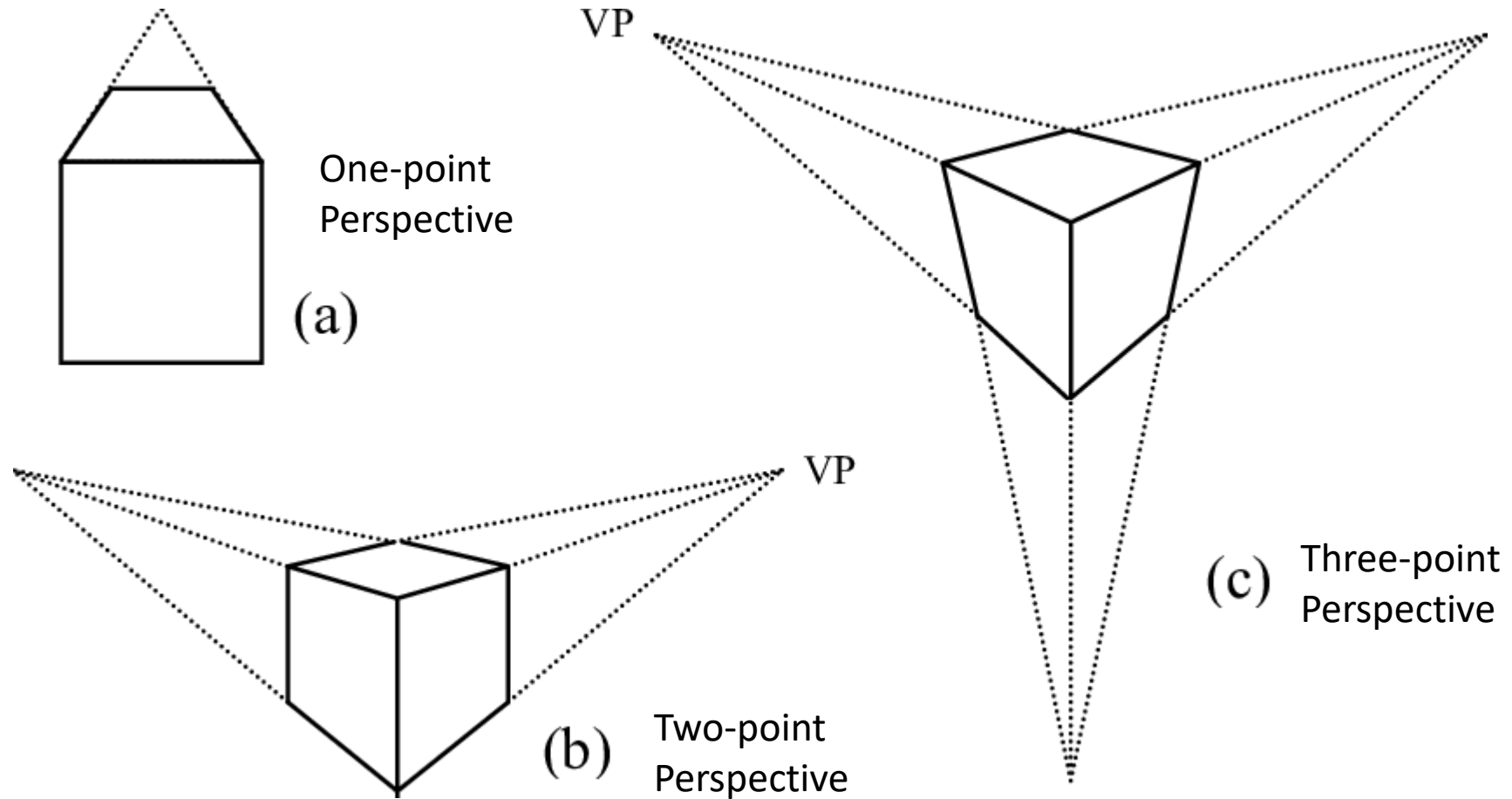


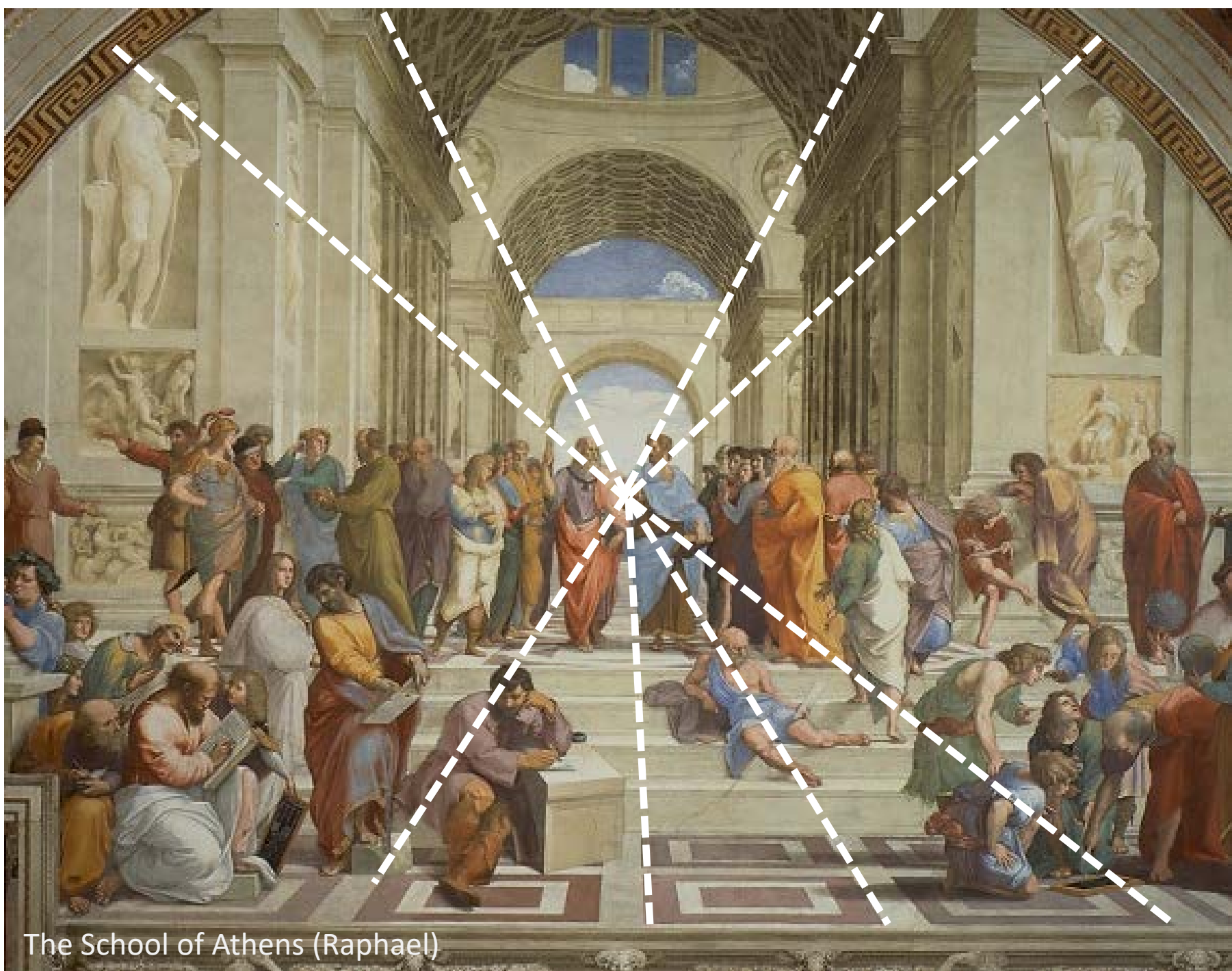
Credit: Vespertunes

Vanishing Points and Horizon Line



Vanishing Points





The School of Athens (Raphael)

How high is the camera?

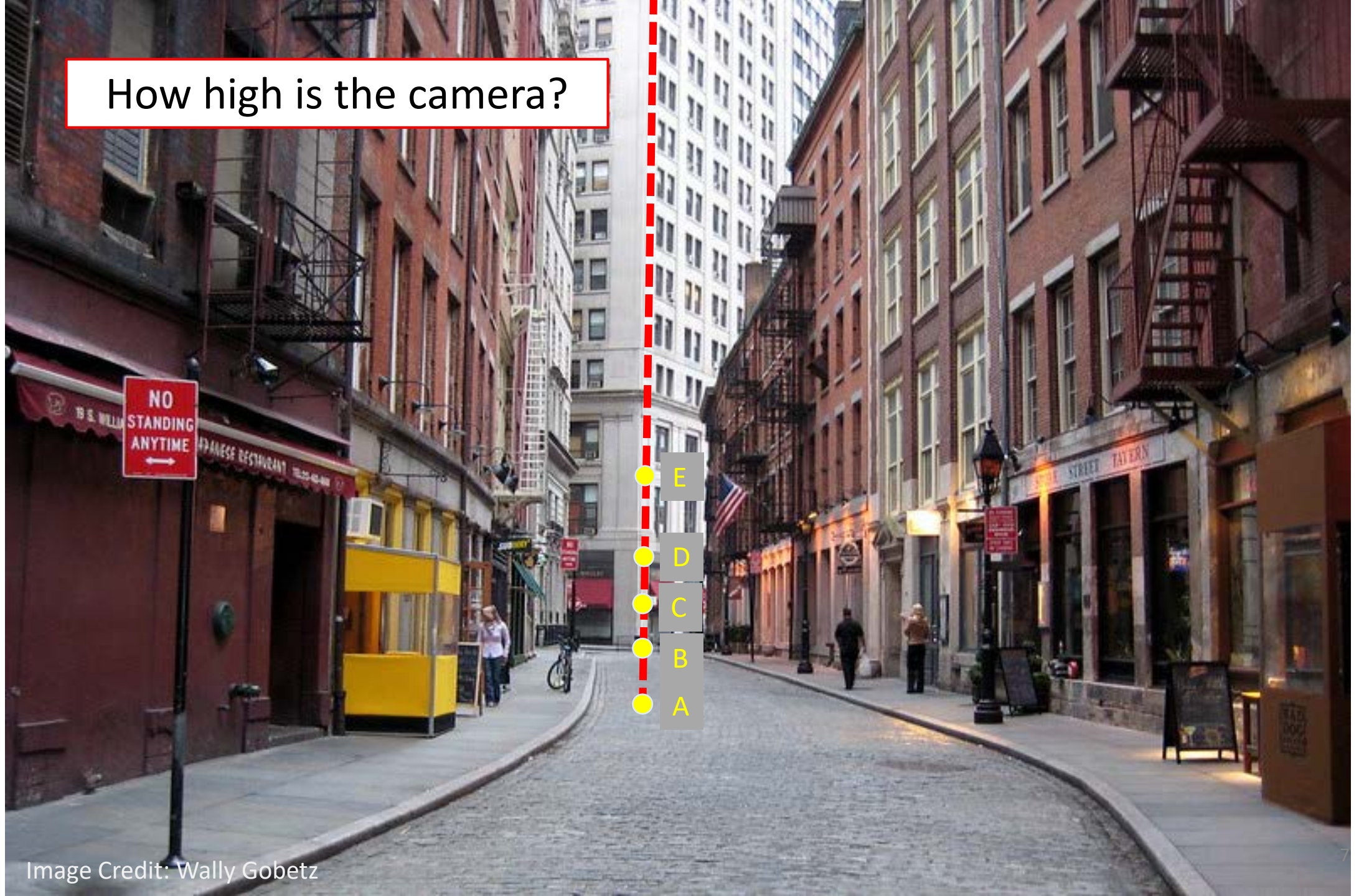


Image Credit: Wally Gobetz



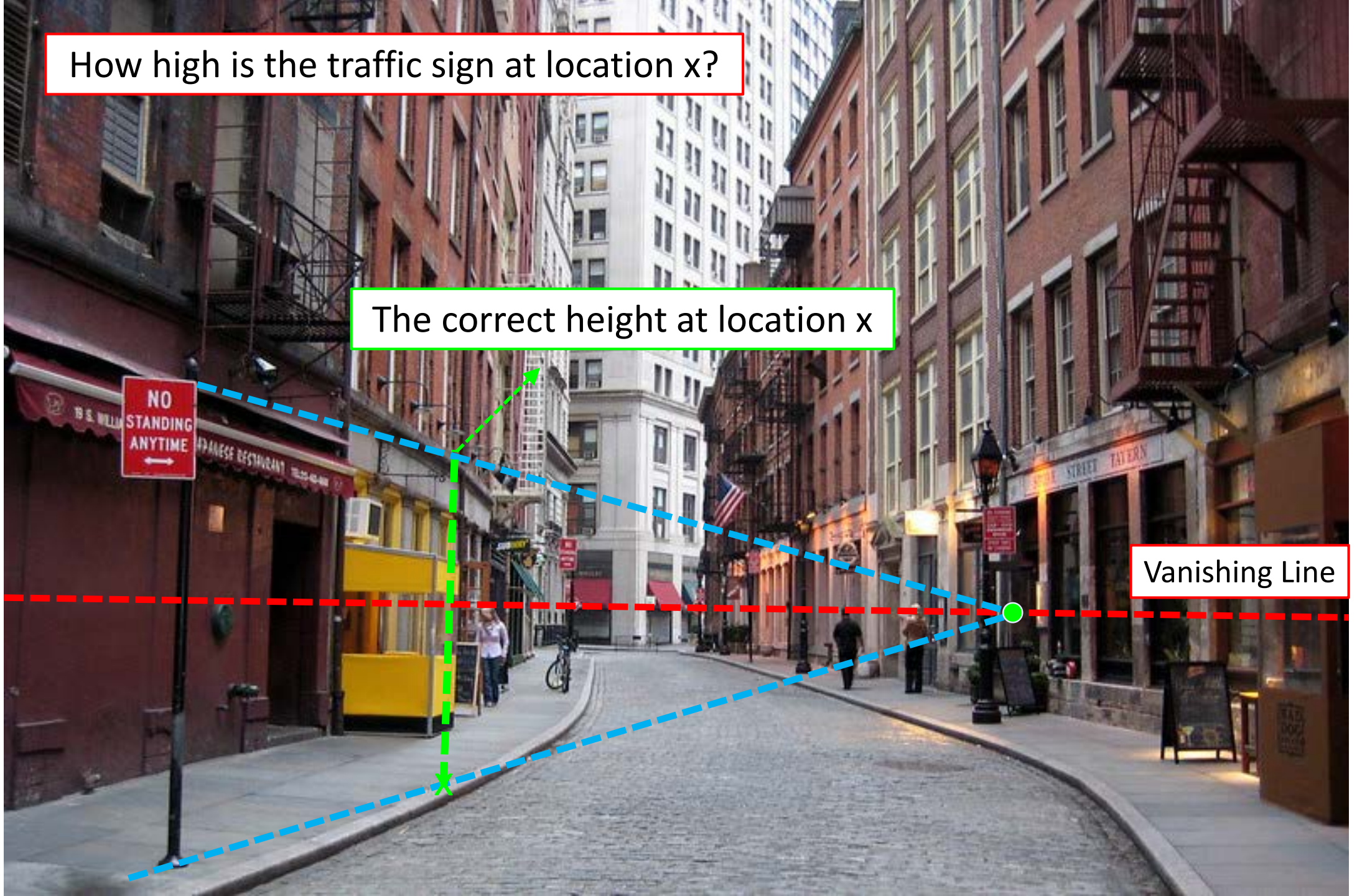
How high is the traffic sign at location x?



How high is the traffic sign at location x?

The correct height at location x

Vanishing Line



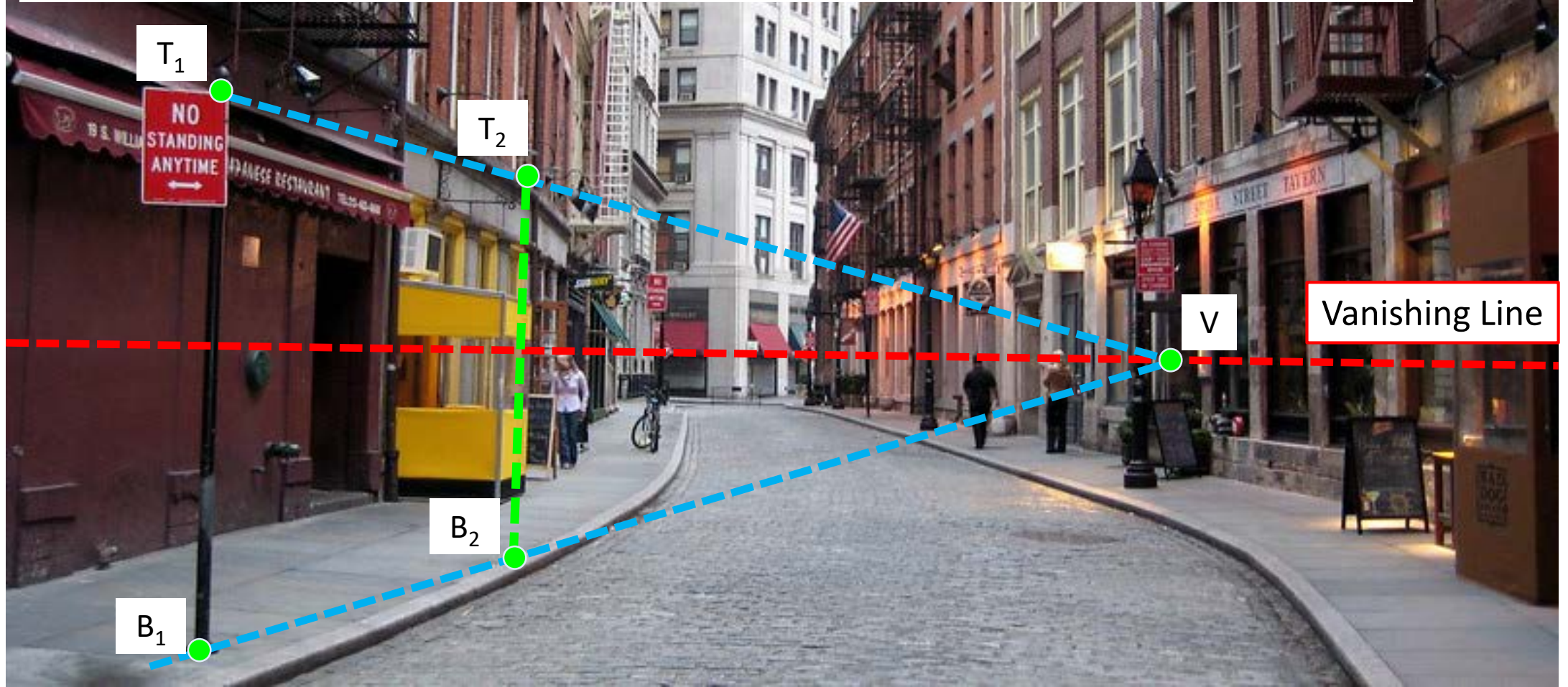
1. Find the vanishing line.

2. Draw the line $\overline{B_1B_2}$ connecting the two bottom points (B_1 and B_2).

3. Suppose $\overline{B_1B_2}$ intercepts the vanishing line at V .

4. Draw the line $\overline{VT_1}$ connecting the top point T_1 and V .

5. Draw a vertical line from B_2 . Then T_2 is the interception of the vertical line and $\overline{VT_1}$.



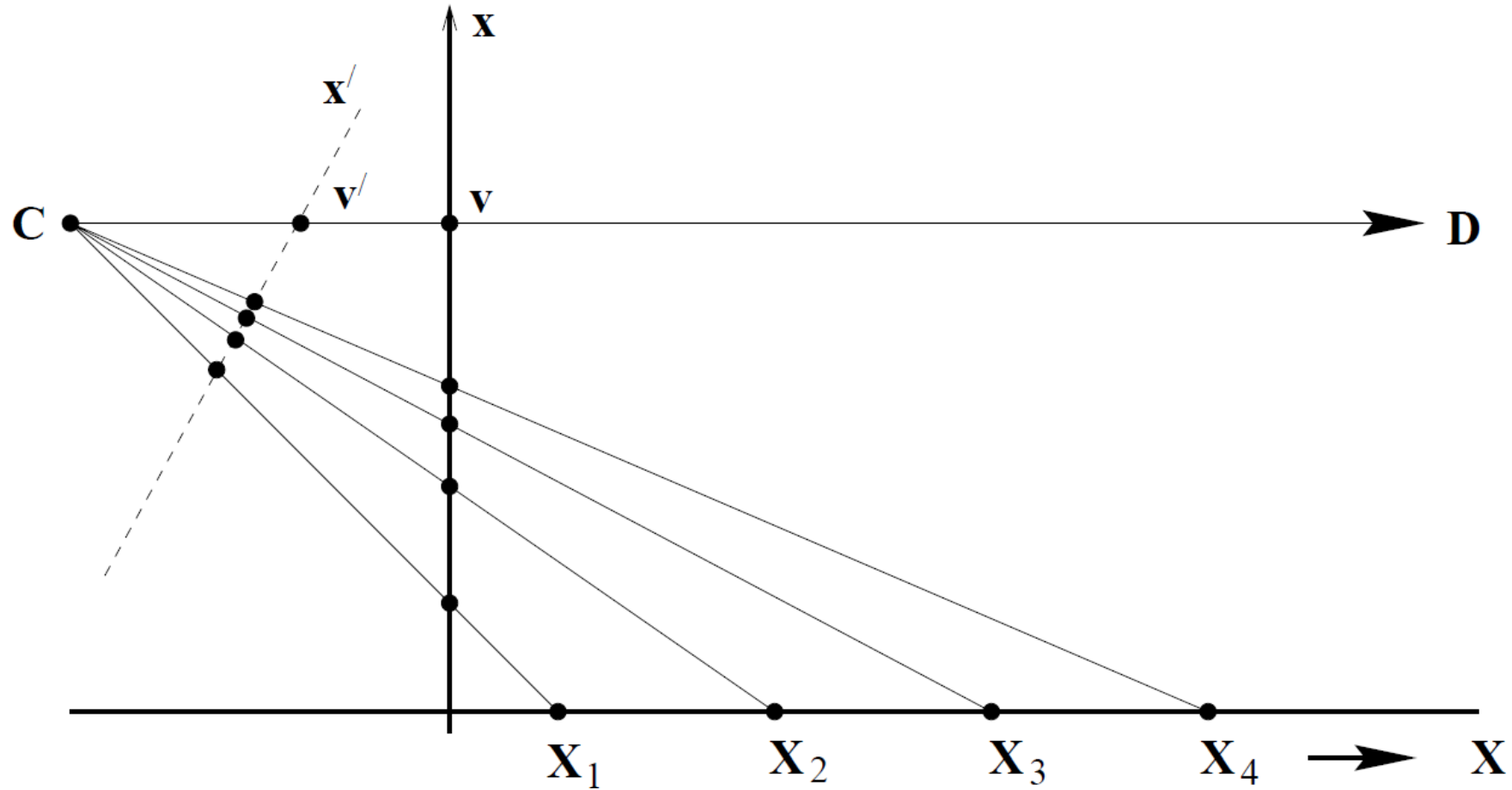


Vanishing Line

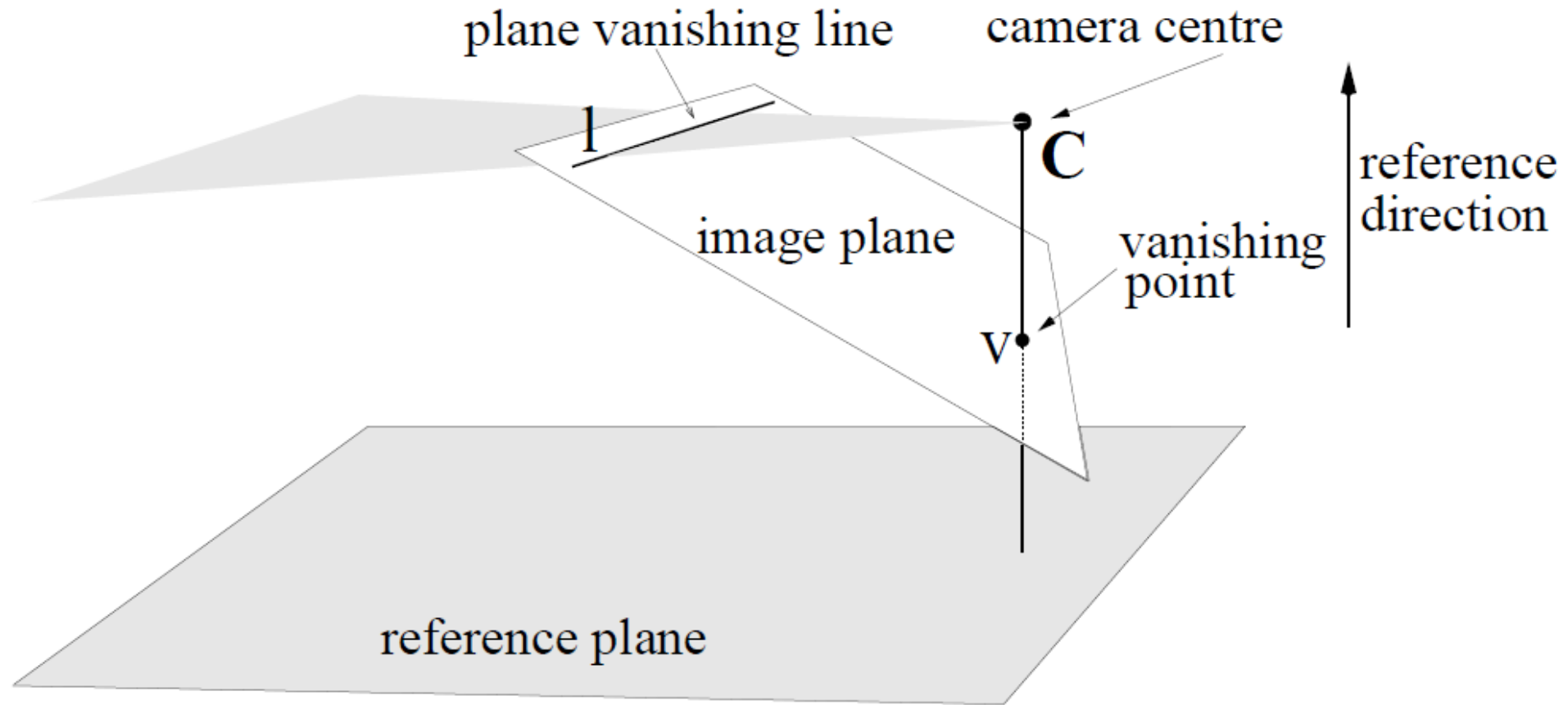
Verification on existing traffic signs.



Vanishing Point Formation



Vanishing Line



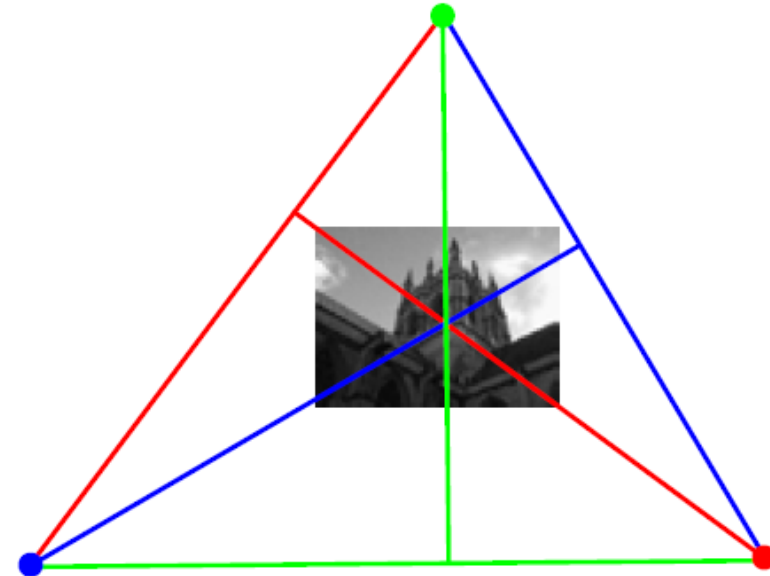
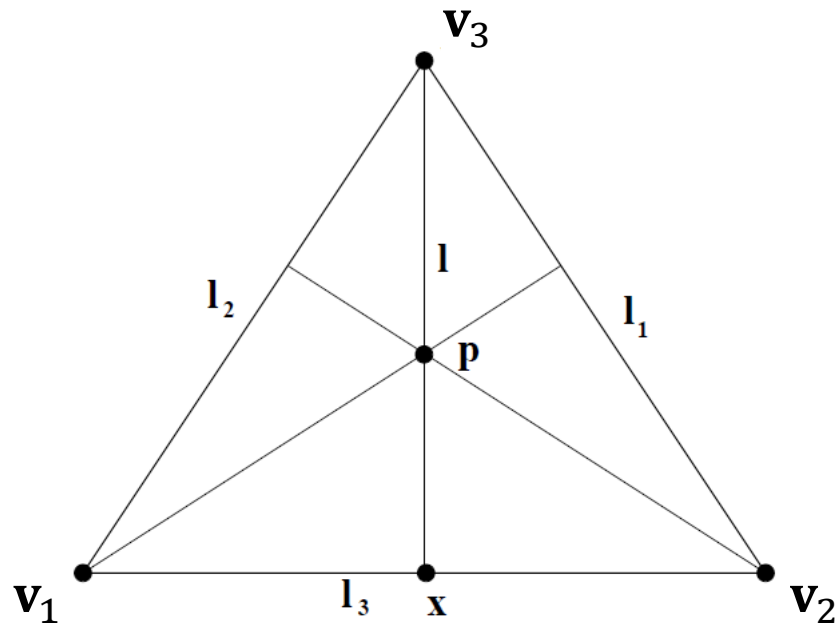


Taken from beach



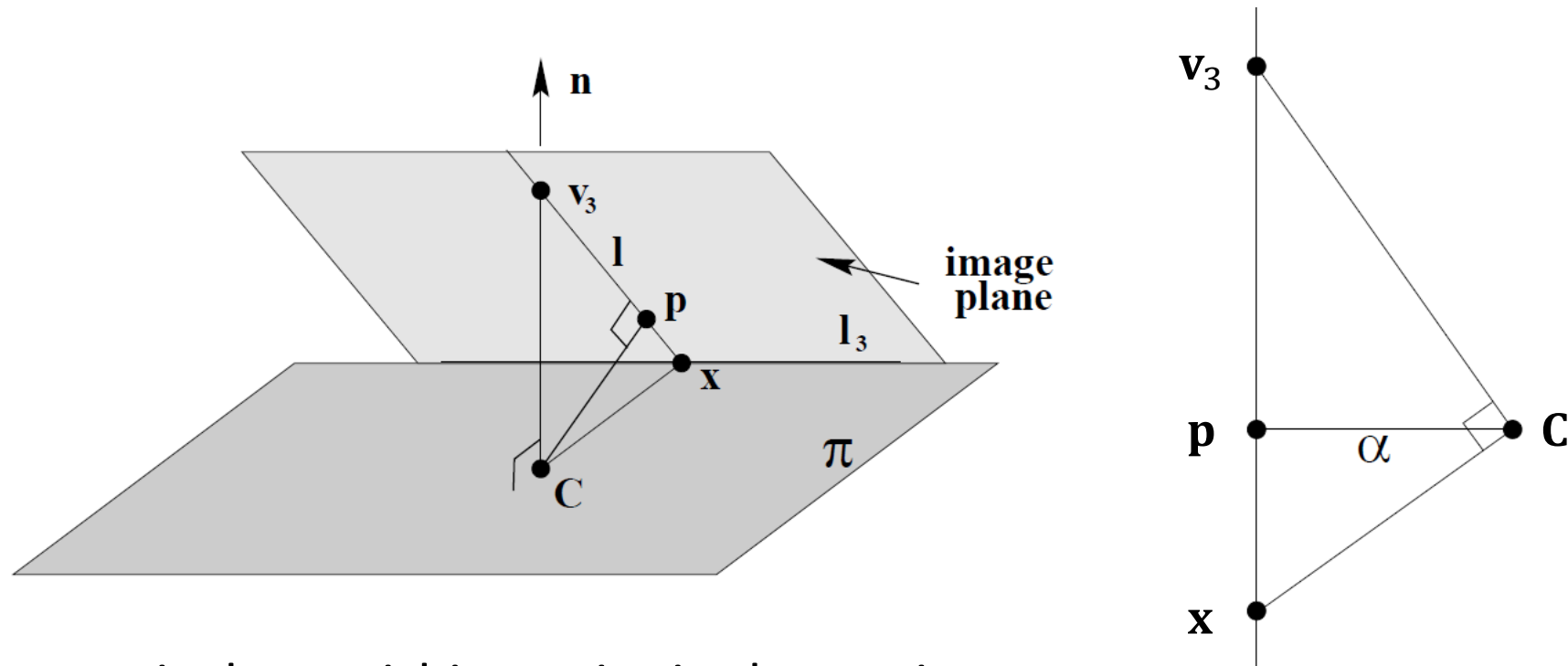
Taken from the 6th floor

Geometric Construction of Principal Point



- v_1, v_2, v_3 are vanishing points.
- The principal point is the orthocenter of the triangle formed by the three vanishing points.
- Assume zero skew and unit aspect ratio.

Geometric Construction of Focal Length



- v_3 is the vanishing point in the z-axis.
- C is the camera center, and p is the principal point.
- α is the focal length, and $\alpha^2 = d(p, v_3) d(p, x)$ by similar triangles.

Outline

- Vanishing Points (VP) and Horizon Line (HL)
- **Detecting VPs and HL**
- Cross-Ratio

Detecting Lines



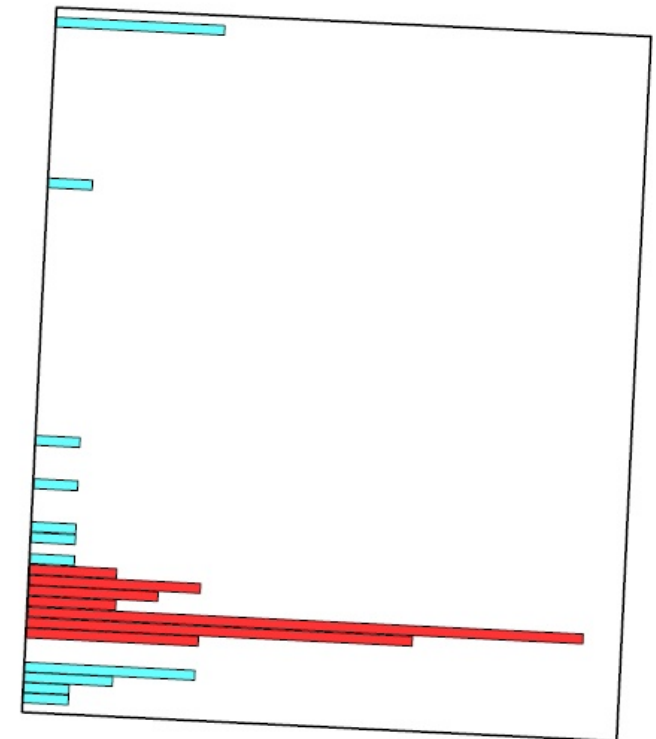
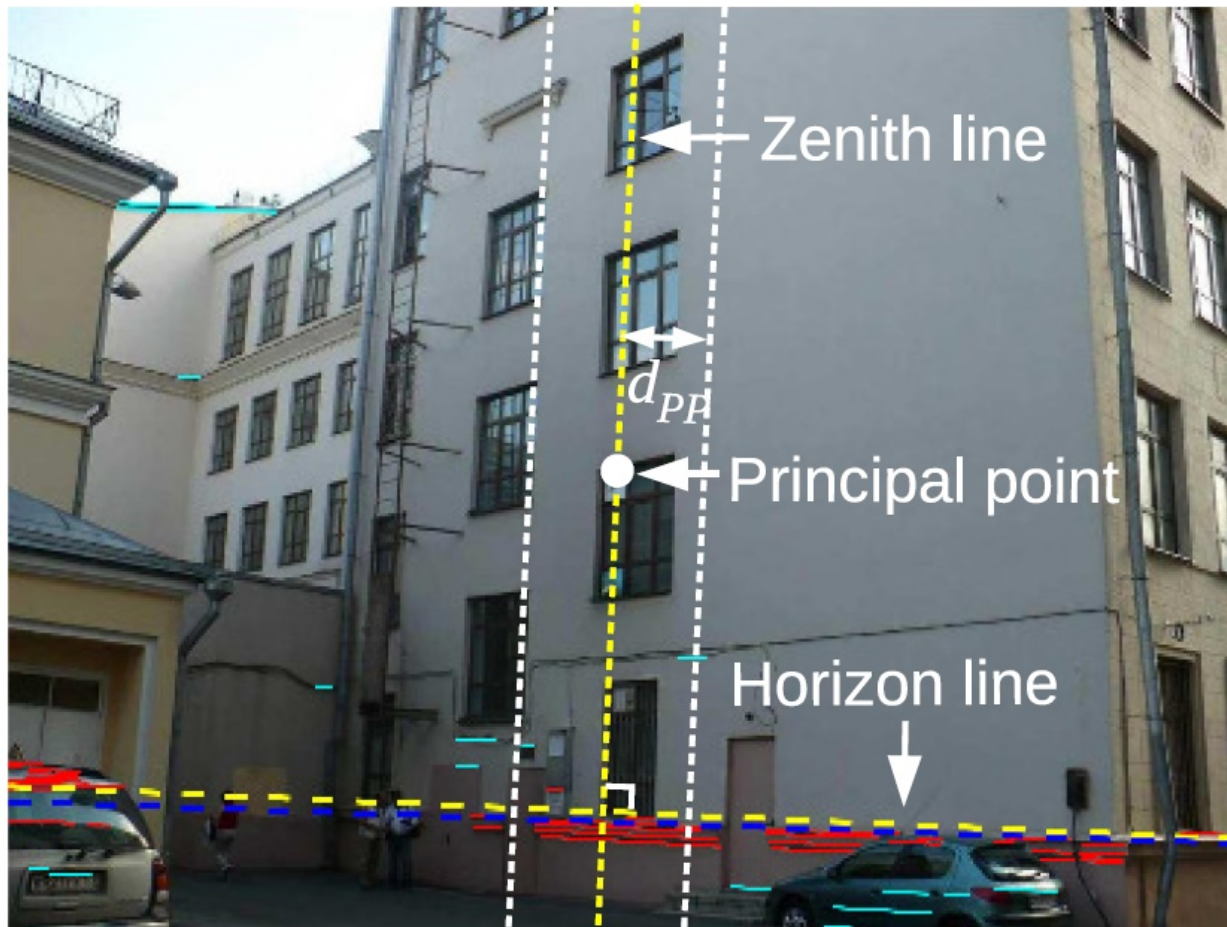
- Line segment grouping yields a large number of VPs.
- One approach is to detect the zenith (vertical VP) and the horizon line (HL).
- Then constraining the horizontal VPs on the HL.

Searching along Zenith Line



- It is difficult to establish a ground truth from selecting hundreds of relevant VPs.
- The ground truth of HL is less subjective than that of VPs.
- The zenith line is the line connecting the principal point and the zenith.

Searching along Zenith Line



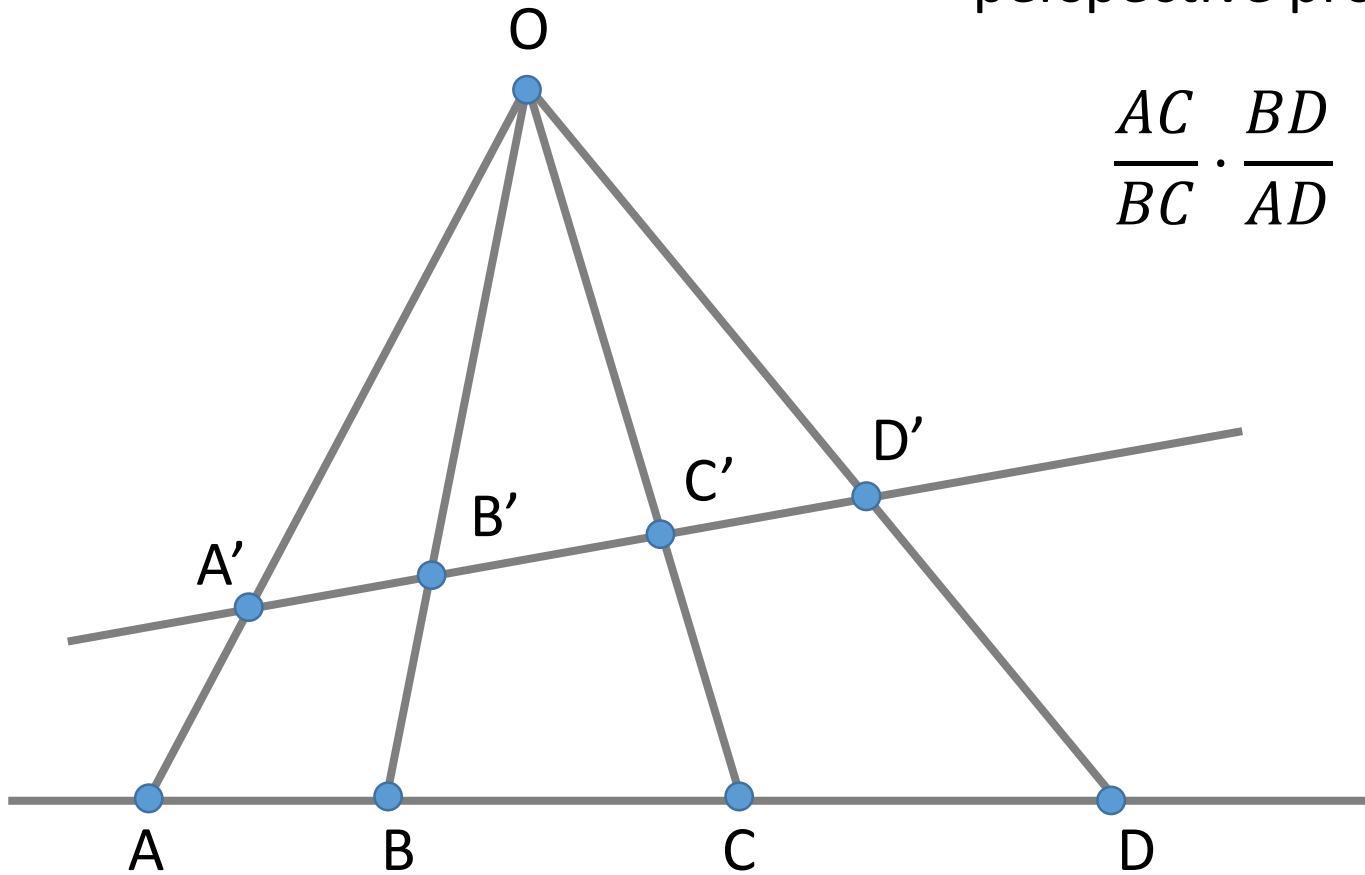
Cross-Ratio

The cross-ratio is invariant under perspective projection.

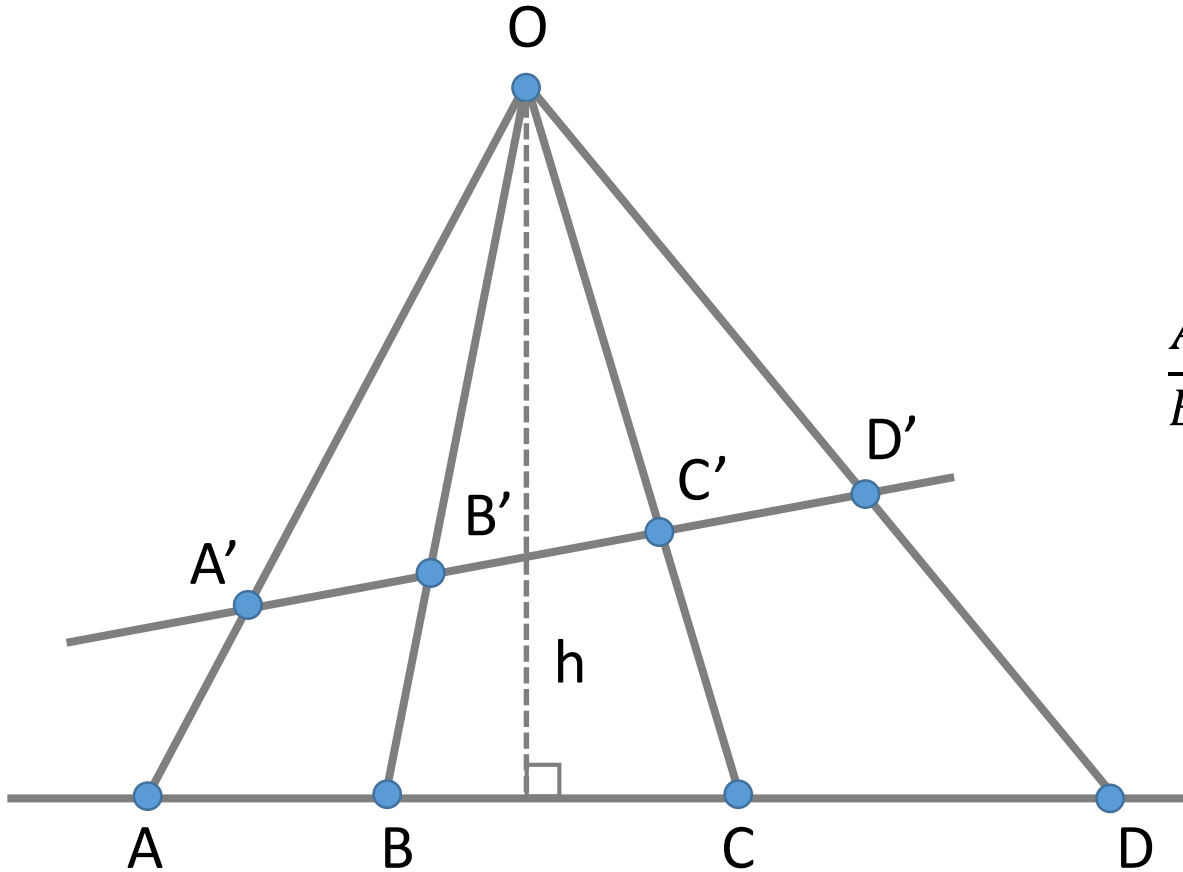
$$\frac{AC}{BC} \cdot \frac{BD}{AD} = \frac{A'C'}{B'C'} \cdot \frac{B'D'}{A'D'}$$

Another version:

$$\frac{AC}{AB} \cdot \frac{BD}{CD} = \frac{A'C'}{A'B'} \cdot \frac{B'D'}{C'D'}$$



Cross-Ratio



$$\text{area } OCA = \frac{1}{2} h \cdot CA = \frac{1}{2} OA \cdot OC \sin \angle COA$$

$$\text{area } OCB = \frac{1}{2} h \cdot CB = \frac{1}{2} OB \cdot OC \sin \angle COB$$

$$\text{area } ODA = \frac{1}{2} h \cdot DA = \frac{1}{2} OA \cdot OD \sin \angle DOA$$

$$\text{area } ODB = \frac{1}{2} h \cdot DB = \frac{1}{2} OB \cdot OD \sin \angle DOB$$

$$\frac{AC}{BC} \cdot \frac{BD}{AD} = \frac{OA \cdot OC \sin \angle COA}{OB \cdot OC \sin \angle COB} \cdot \frac{OB \cdot OD \sin \angle DOB}{OA \cdot OD \sin \angle DOA}$$

$$= \frac{\sin \angle COA}{\sin \angle COB} \cdot \frac{\sin \angle DOB}{\sin \angle DOA} = \frac{\sin \angle C'OA'}{\sin \angle C'OB'} \cdot \frac{\sin \angle D'OB'}{\sin \angle D'OA'}$$

$$= \frac{OA' \cdot OC' \sin \angle COA}{OB' \cdot OC' \sin \angle COB} \cdot \frac{OB' \cdot OD' \sin \angle DOB}{OA' \cdot OD' \sin \angle DOA}$$

$$= \frac{A'C'}{B'C'} \cdot \frac{B'D'}{A'D'}$$

(1)

$$\frac{AC \times BD}{BC \times AD} = \frac{A'C' \times B'D'}{B'C' \times A'D'}$$

$$\frac{(30 + 20) \times (20 + 10)}{20 \times (30 + 20 + 10)} = \frac{(7 + W)(W + 6)}{W(7 + W + 6)}$$

$$5W(W + 13) = 4(W + 7)(W + 6)$$

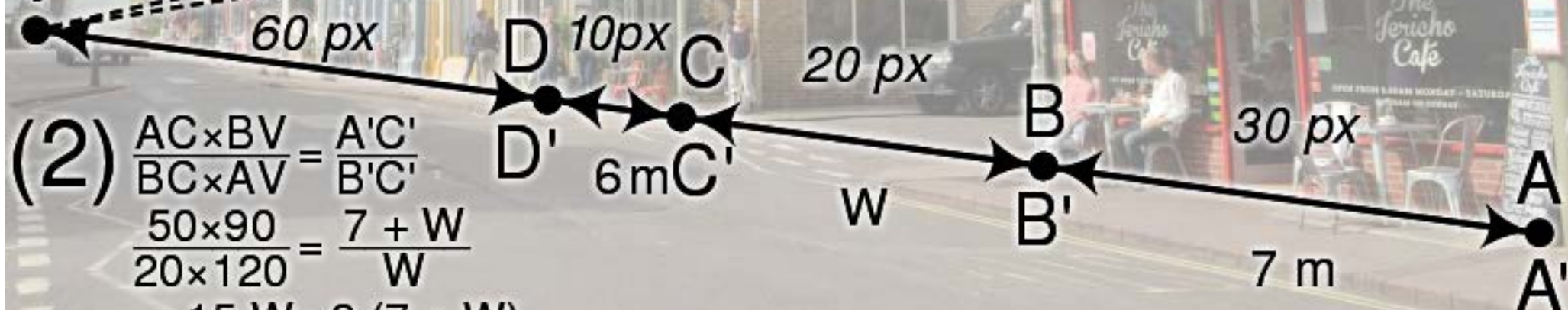
$$5W^2 + 65W = 4W^2 + 52W + 168$$

$$W^2 + 13W - 168 = 0$$

$$(W + 21)(W - 8) = 0$$

$$W > 0 \therefore W = 8 \text{ m}$$

V



(2)

$$\frac{AC \times BV}{BC \times AV} = \frac{A'C' \times B'V'}{B'C' \times A'V'}$$

$$\frac{50 \times 90}{20 \times 120} = \frac{7 + W}{W}$$

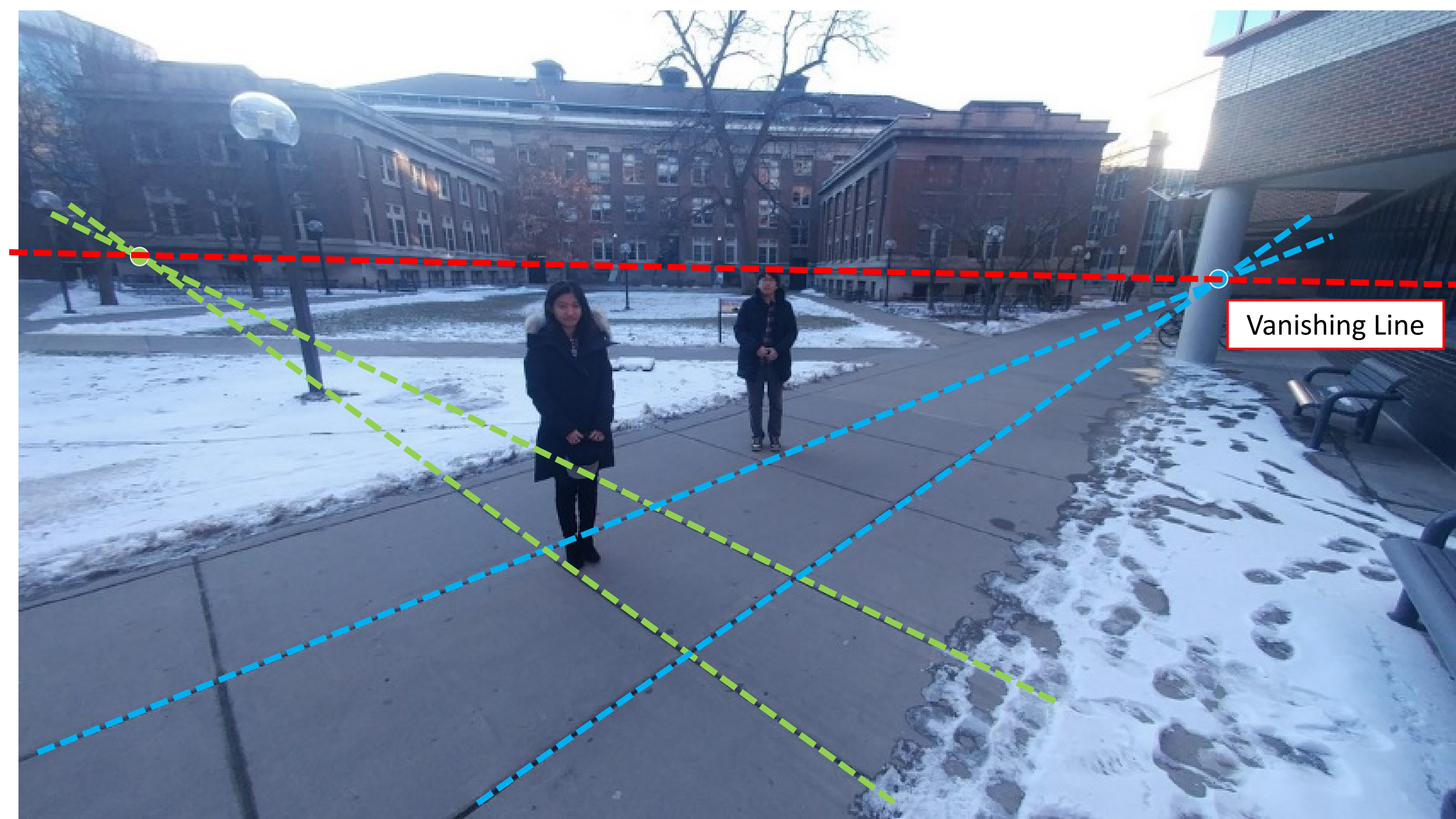
$$15W = 8(7 + W)$$

$$7W = 56 \therefore W = 8 \text{ m} \quad [\text{Credit: cmglee}]$$

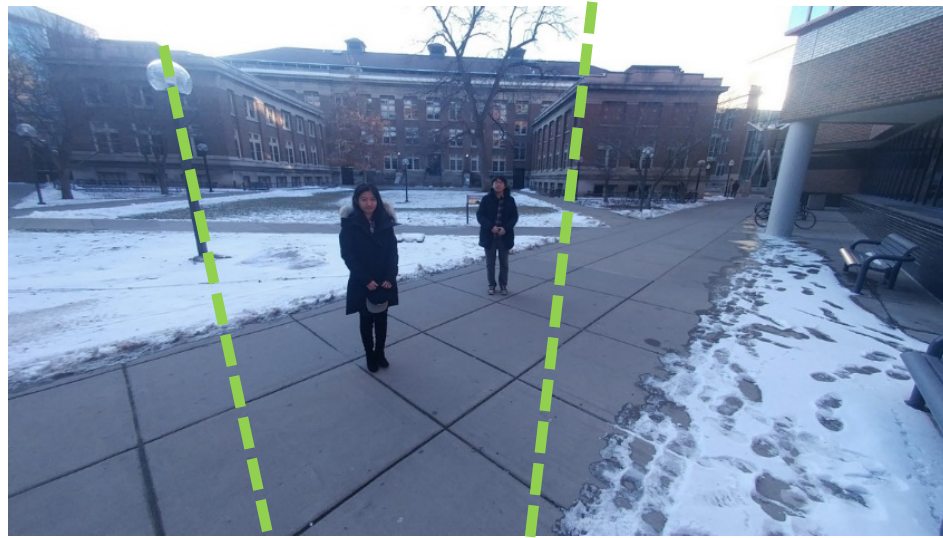
How high is the male student?

1.6m

???



Vanishing Line



Vanishing Point

A: The head point of the male student at the location of the female student

B: The head point of the female student

C: The foot point of the female student

A

B

C

D

Vanishing Line

$$\frac{AC}{BC} \cdot \frac{BD}{AD} = \frac{A'C'}{B'C'} \cdot \frac{B'D'}{A'D'}$$

$$\frac{A'C'}{B'C'} \cdot \frac{B'D'}{A'D'} = \frac{H}{1.6} \cdot \frac{\infty}{\infty} = \frac{H}{1.6}$$

H is the height we want to compute

AC, BD, BC, AD can be directly measured from the image

$$H = 1.6 \cdot \frac{AC}{BC} \cdot \frac{BD}{AD}$$

Vanishing Point